

EXP 8: Write a program to implement the Linear Regression (LR) algorithm in Python

AIM

To implement the Linear Regression (LR) algorithm in Python and predict the output using a suitable dataset.

SOFTWARE USED

- Google Colab
- Python 3.x
- Pandas Library

PROCEDURE

1. Import the required libraries.
2. Create a sample dataset.
3. Split the dataset into training and testing sets.
4. Train the Linear Regression model.
5. Predict the output for the test data.
6. Calculate and display the model accuracy.

GOOGLE COLAB CODE

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
import numpy as np

# Sample Dataset
X = np.array([[1], [2], [3], [4], [5], [6], [7], [8], [9], [10]])
y = np.array([2, 4, 6, 8, 10, 12, 14, 16, 18, 20])

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Create Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)
print("Predicted Values:")
print(y_pred)
print("\nActual Values:")
print(y_test)

# Accuracy
accuracy = r2_score(y_test, y_pred)
print("\nR2 Score:", accuracy)

# Predict a new value
new_value = [[12]]
prediction = model.predict(new_value)
print("\nPrediction for X = 12:")
print(prediction[0])
```

OUTPUT

Predicted Values:

[18. 4.]

Actual Values:

[18 4]

R2 Score: 1.0

Prediction for X = 12:

24.000000000000007

RESULT

Thus, the Linear Regression (LR) algorithm was successfully implemented in Python. The model was trained and tested using a sample dataset, achieving an R^2 score of 1.0, and it correctly predicted the output for a new input value.